

CSE/ECE470: Computer Systems Architecture

Class Overview

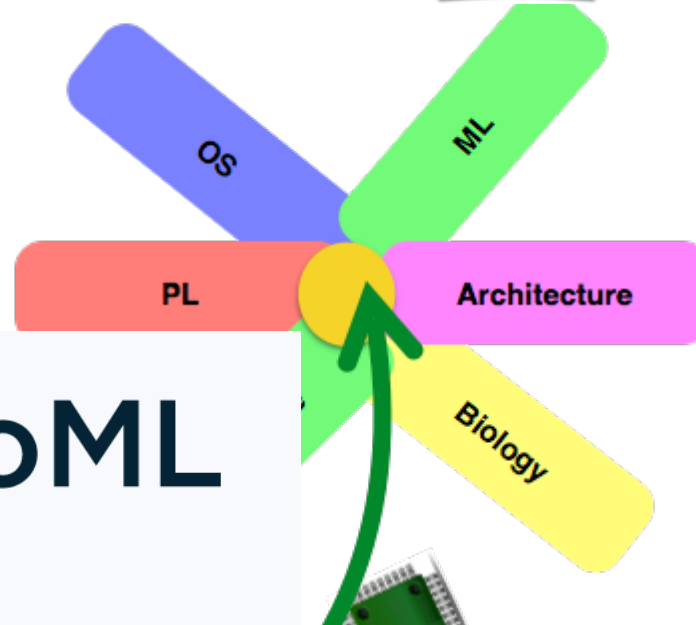
Spring 2022

Luis Ceze



Eating is more important than nutrition :)

Machine learning compilation and optimization



~Approximate computing

Efficiency and performance

Output quality-of-result



OctoML

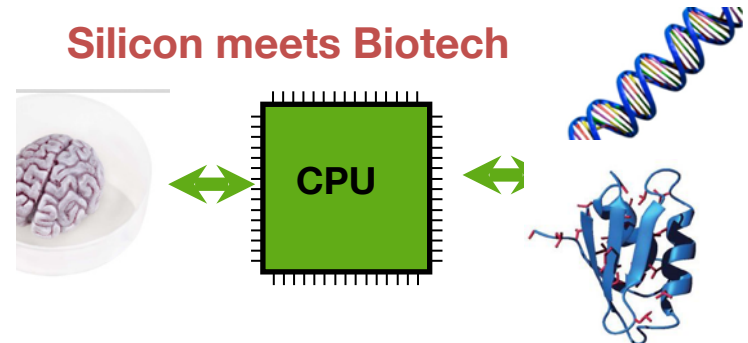
Machine learning on your machine learning.

From the creators of [Apache TVM](#), [XGBoost](#) and [Apache MxNet](#), OctoML brings the cutting edge of AI, Systems, programming languages, compilers and architecture to make machine learning systems easier to optimize and deploy.

[Find out more](#)



Silicon meets Biotech



TeAm



My research mainly focuses on efficient machine learning algorithms and systems. Outside of research, I enjoy being outside. My favorite activities include tennis, hiking and skiing.

Chien-Yu Lin: 3th year, advised by Luis Ceze



Astitwa S. Lathe (TA)

Background

- Second year MSEE student.
- Concentration: Embedded and VLSI.
- Worked as a Systems Engineer with research and publication experience.

Interests

- Research: Ubiquitous Computing, Reconfigurable Computing.
- Sports/Games: Badminton, Golf, Pool, Chess.
- Others: Reading (Novels, Mangas, Light novel).

Contact

- Email: aslathe@uw.edu.

Who are you? 😊

- Diverse, yay! ~45 of you
- Quick round of introductions
 - Interests
 - Other classes
 - *Anything* else you want us to know about you!
- This class will be *much* better if we have rich discussions. So don't be shy!

Why Learn Computer Architecture?

- All your programs **run in real machines**
 - you should understand well how they work
 - (you know..., ML needs **M**achines ☺)
- Computer architecture is at a crossroads
 - **multi-cores, FGPA's, GPU's** everywhere
 - **power, dependability** are major concerns
 - **technology trends** motivate new applications
 - **security issues** thrive with complexity

Class Outline

- Intro, Basic Principles, Technology and Metrics
- The Hardware/Software Interface (ISA)
- Superscalar/OoO execution
 - And security issues in modern CPUs (e.g., Spectre/Meltdown)
- Memory Hierarchy
- Parallelism (ILP, DLP, TLP)
 - Multiprocessors (Coherence/Consistency)
 - Multithreading/GPUs
- Specialized computing (FPGAs, ASICs)
 - Emphasis on machine learning
 - Warehouse-scale computing
- Silicon technology trends (e.g., 3D integration) and impact in IT
- Putting it all together with Real HW Systems dissections
 - Modern microprocessors, Xbox One, Raspberry Pi, Hololens, Tesla FSD chip, Cerebras?

Administrivia

- Readings:
 - Book (Hennessy and Patterson, 6th edition)
 - Readings for your own benefit, slides often enough
 - Papers: **Readings essay assignments**
 - write 1-2 (short ☺) **paragraphs** about the key points, submit to dropbox
- **Homework: 5** (some with light coding)
- Grading:
 - 20% reading assignments
 - 80% homeworks
- Check our website regularly
 - Shared calendar with classes, office hours, project meeting slots etc
 - `https://courses.cs.washington.edu/courses/cse470/sp22`

Website up